Achernar

Unified Astronomy Imaging and Control System

System Architecture v6

Public-Data-First Model Development,
Compute-Aware Runtime, and Bounded Control

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Purpose

Achernar should use the strongest validated perception model that a local machine can execute reliably within its operational budget. Small models form the minimum deployment tier. More capable hardware can qualify higher-capacity variants without changing the physical meaning of observations or the authority of the control and safety layers. This revision extends the v5 architectural restatement of v4 with an original-source public-dataset audit and a revised local collection strategy. Inherited implementation descriptions are the v5 baseline except where this revision explicitly rechecks the affected source contracts. It is not a byte-for-byte recovery of the unavailable v4 LaTeX source. Proposed APIs, configuration, model tiers, and hardware support are identified as design work; this document does not certify implementation or measured performance.

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1 Executive Summary

Achernar combines an on-site Rust daemon, a React operator interface, a Tauri desktop shell, and a Django cloud platform. Hardware access, local acquisition, measurement, and safety decisions belong to the daemon. Cloud services support archiving, coordination, training, and analysis where implemented. The local observing path must remain useful without a network connection.

V6 retains v5's compute-aware runtime for scalable perception and changes the data strategy to public-data-first experiments with targeted local evidence. The runtime inventories installed artifacts, identifies the execution providers available in its build, validates deployed behavior, benchmarks eligible candidates, and chooses a qualified implementation for each model family. Hardware discovery supplies an initial prior; quality evidence and measured resource constraints determine the selection.

1.1 Scope and design status

The words *must* and *shall* in proposed sections specify acceptance requirements for future implementation. They do not imply that a module, endpoint, or trained model already exists. Implementation descriptions carried from v5 retain their October 3 baseline; affected data/measurement contracts were rechecked on October 12. This is not a fresh end-to-end hardware certification. Dated audits remain historical records, and the source code and migrations determine the current executable contract.

Area	Status at this revision
Classical vision	Implemented HFR, background, source rejection, and trailing measurements. Unknown physical quantities remain unavailable where they cannot be measured honestly.
FQE inference	Implemented optional shadow ONNX path, 64-pixel star crops, batched execution, robust pooling, and hash-bound calibration. No accelerator provider is configured by this path.
Shipped FQE artifact	Random reference graph for integration testing; `runtime_eligible=false`. It is not a production model.
COC and TED neural paths	Metadata/placeholders exist; the runtime uses heuristic/classical behavior. No trained production ONNX artifacts ship.
Focus drift	Implemented three-state Kalman estimator, with advisory output and per-rig mutable state. It is not an ONNX network.
Generic model runtime	Registry scaffolding is not wired into startup; generic ONNX runner is unfinished. Variant selection, provider qualification, benchmark cache, and runtime status endpoints are proposed.
Accelerator releases	CUDA/TensorRT, CoreML, DirectML, and OpenVINO are candidate deployment paths, not a validated support matrix.

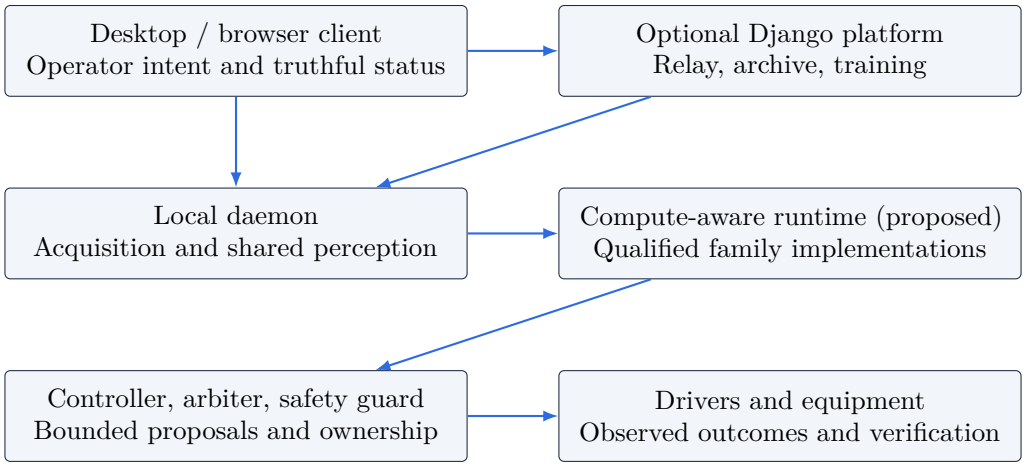
1.2 Target users and deployment principle

The same daemon architecture serves field equipment, backyard rigs, personal laptops, and observatory workstations. A computer with more available compute may qualify a larger or higher-resolution perception model. Executing unchanged weights faster does not improve those weights' prediction quality. Additional quality must come from a trained and validated variant, additional useful observations, or a validated ensemble.

2 System Architecture Overview

2.1 Architecture layers

The equipment layer contains cameras, mounts, focusers, wheels, and sensors. The daemon owns device connections, acquisition, local storage, shared perception, arbitration, and control. The client layer displays authoritative daemon state and sends commands through a documented transport. The cloud layer provides remote coordination and optional data services. External weather, catalog, and astrometric resources require explicit freshness and availability handling.



The drawing shows responsibilities, not independent authority. A model runtime never sends motor, exposure, filter, or mount commands. A larger model supplies a potentially better estimate through the same family contract.

2.2 One product, several qualified deployments

Perception families can use different tiers on one machine. For example, FQE may run a larger network while COC remains on a small network and the background estimator remains classical. Per-family selection is subject to a joint resource budget: qualifying each family in isolation is insufficient if their simultaneous memory or compute demand exceeds the observing budget.

3 Equipment Layer

3.1 Capabilities and hardware qualification

The design retains the v4 camera, mount, focuser, filter-wheel, and environmental sensor categories, including OLS/vendor SDK paths, Alpaca, and planned INDI or direct interfaces. A declared Rust driver enum or an architectural diagram is not evidence that a device has a working driver. Publish capability availability, connection state, and disabled reasons from the actual implementation.

Per-rig profiles hold measured travel bounds, backlash, settling time, optics, pixel size, filter offsets, and sensor calibration. Unknown position, temperature, tracking, or pointing is represented as unknown. Do not synthesize zero positions or safe states to fill a UI or a model feature vector.

3.2 Compute and equipment are separate capabilities

An accelerator can improve image analysis without adding support for a camera or motor. Device access must be tested separately from inference provider support. Release records identify OS, architecture, native dependencies, driver/SDK versions, and actual equipment exercised.

4 Achernar Daemon: Rust and C++ Core

4.1 Hardware abstraction

Rust coordinates asynchronous work and exposes typed device operations. The C/C++ bridge and supported OLS drivers handle the native hardware interfaces. Ownership and movement validation remain between action proposals and drivers. Recovery must distinguish disconnection, unsupported capability, timeout, and an unknown result after a partially completed command.

4.2 Image processing engine

Acquisition, calibration, source measurement, preview rendering, and optional stacking have different data and timing requirements. Preserve raw science assets and record the lineage of derived previews and measurements. A display stretch must not become the input to physical flux measurement unless that input is explicitly part of a validated analysis contract.

Compute reservations include debayering, encoding, stacking, storage, and camera I/O. Accelerator memory reported as free at one instant is not an operational allocation guarantee. The runtime must leave configured headroom for these other consumers and the operating system.

4.3 Local AI/ML processing

v6 requirement; implementation pending. Agents request a semantic model family. A runtime handle binds the selected artifact, provider, preprocessing, postprocessing, and calibration. Hardware-specific code lives below this handle. Classical algorithms remain available when a learned candidate is absent, unqualified, or unhealthy.

Current implementation. FQE creates its own ORT session in `achernar-daemon/src/vision/fqe_runner.rs`; application state stores a single `fqe_runner`. The new runtime will replace this wiring in a later code change. The generic `OnnxRunner` is currently a stub.

4.4 Control systems

Sequence control, classical autofocus, drift advice, exposure policies, and future filter/mount agents consume observations with explicit freshness, validity, and units. Motion and exposure decisions retain bounded travel, ownership locks, settle periods, and outcome verification. Resource selection does not change an agent's autonomy mode.

4.5 Local storage

SQLite migrations define executable schemas; immutable frame assets preserve acquisition provenance. Mutable rig state, runtime benchmarks, provider engine caches, and selection history

belong in the configured data directory. Shipped model bundles are immutable and must not accumulate nightly learning or local device measurements.

4.6 Daemon server

Current implementation. Axum exposes local HTTP routes under `/api/v1`, telemetry under `/ws/telemetry`, and a health endpoint. The default bind address is loopback. The current router does not expose a generic AI runtime API. Broader network binding must be assessed with the authentication and transport design; a configuration example is not evidence of remote authorization enforcement.

5 Interfaces, Data Contracts, and Reliability

5.1 API specifications

Current route definitions are in `achernar-daemon/src/main.rs`. The command registry is the authority for commands exposed through the relay. Availability in a local router does not automatically imply relay availability.

Route or surface	Contract and status
`GET /api/v1/status`	Current equipment and observation status.
`GET /api/v1/commands`	Current published relay command inventory.
`GET /api/v1/drivers`	Current driver capability reporting.
`POST /api/v1/camera/capture`	Current acquisition request; physical capture provenance is separate from analysis-only snapshots.
`/api/v1/camera/stream`	Current binary preview WebSocket. The separate `/stream/mjpeg` route returns a placeholder, not multipart MJPEG.
`/api/v1/focus/*`	Current focus run, abort, status, history, and drift routes; Section 7.5 distinguishes future extensions.
`GET /api/v1/ai/models`	Proposed inventory with artifact and qualification states, including ineligibility reasons.
`GET /api/v1/ai/runtime`	Proposed active candidates, provider placement, resource budgets, baseline, and health.

Administrative rebenchmarking is a future explicit operation. It must not run expensive experiments inside a critical capture or focus transaction and needs authorization, rate limiting, cancellation semantics, and an audit event before it is exposed remotely.

5.2 Core data and provenance

Existing migration families cover sessions, equipment, focus runs, filter/mount records, frame assets, focuser scale, environment logs, and grouping keys. New runtime records will extend

these schemas through migrations; this document does not provide replacement SQL to apply to an existing database.

Every scientific observation must be traceable to the acquired frame or frame generation that produced it. Proposed runtime provenance adds family, variant, artifact digest, preprocessing/postprocessing identifiers, provider and options, actual provider placement, precision, input profile, batch size, calibration, selection epoch, and latency. Keep these fields separate from the scientific value so downstream control remains independent of implementation details.

5.3 Errors and recovery

Model absence or initialization failure degrades to a qualified baseline. Corrupt artifacts are quarantined. Non-finite predictions, mismatched tensor contracts, stale observations, and invalid uncertainty must not produce an apparently valid measurement. Failure emits a reason and a null/unavailable prediction; it does not fabricate zero HFR, zero uncertainty, or certainty of clear sky.

A timeout on an asynchronous wrapper does not necessarily cancel native inference. Bound the worker queue, prevent overlapping abandoned jobs, and discard late results by frame ID and selection epoch. If a provider cannot be interrupted, reserve enough resources for recovery or isolate it in a restartable worker. Do not block hardware abort paths behind a model session lock.

5.4 Configuration

Current implementation. The operational example is `achernar-daemon/achernar.example.toml`. Its current parser has no top-level `[ai]` section. Focus model paths live under `[focus.models]`; drift tuning lives under `[focus.drift]`.

The following is a proposed configuration contract, not runnable configuration:

```
[ai]
model_selection = "auto"
compute_budget = "balanced"  # efficient | balanced | quality

[ai.focus_quality]
model = "auto"              # future explicit variant pin also permitted
# Initial rollout remains shadow-only under existing control authority.
```

Policies adjust resource preferences within hard limits. `quality` cannot bypass promotion, memory reserves, validity, or deadlines. Pins select among qualified candidates; they cannot turn reference weights into an autonomous controller. Separate documented development-only explicit-path loading from the production selection policy.

6 AI/ML System Design

6.1 Design principles

Preserve raw science data; attach uncertainty to estimates with a stated meaning; retain classical operation; expose failure reasons; evaluate by rig and observing condition; keep control authority independent of model capacity. Local learning and feedback must retain provenance and must not silently replace promoted artifacts during an observing session.

6.2 Live image intelligence

6.2.1 Focus quality

FQE estimates physical HFR and predictive uncertainty. Current inputs are normalized per-star crops and outputs are pooled in log-HFR space. Do not reintroduce the uncalibrated reciprocal `sharpness_score` used by older examples. A larger crop, higher input resolution, or ensemble is a new scientific variant and requires evidence on the same evaluation population.

6.2.2 Background and SNR

The classical estimator is always available. A future neural path may estimate noise maps or quality, but must not silently denoise raw images used for scientific measurements. Calibration of gain, read noise, black level, and saturation remains necessary regardless of GPU capacity.

6.2.3 Trailing and elongation

Classical source shape measurements provide the baseline. A future small CNN may resolve ambiguous cases; larger models are justified only when their incremental scientific benefit is demonstrated. Elongation direction in image coordinates is not automatically a mount RA/Dec diagnosis without a validated coordinate mapping and adequate supporting telemetry.

6.2.4 Cloud and obstruction

COC is a candidate scalable family with a strongly data-limited quality ceiling. Thin cirrus, moonlight, dew, contrails, gradients, and partial obstruction need representative labels. Current heuristic cloud indicators are not a promoted learned classifier or an independent weather interlock.

6.2.5 Focus drift

FDF remains a CPU state estimator. It learns rig-specific drift from verified focus outcomes and temperature observations. Its quality depends on calibration, temperature coverage, and time history rather than a larger GPU.

6.3 Settings and exposure recommendation

Rules and small contextual bandits can propose exposure/gain actions after appropriate data collection and policy evaluation. Updates happen at permitted sequence boundaries. Their

implementation remains fixed across hardware tiers; their per-rig state may still adapt over time.

6.4 Planning and target ranking

Visibility, horizon, moon separation, target integration requirements, and switching costs feed deterministic scheduling or optimization. Planning can run locally or in a cloud service, but an offline cached plan must remain interpretable and subject to local safety checks.

6.5 Post-capture analytics

Quality review, anomaly detection, retrieval, and archive analysis are separate from live actuation. Record whether a quality label is a classical measurement, model estimate, operator label, or composite policy result. A model should not validate itself against its own automatically accepted outputs.

6.6 Tri-modal embedding and AstroM3Edge

AstroM3 integration and the proposed image/telemetry edge retrofit are distinct deployments. The actual model configuration determines embedding dimension and modality contracts; the historical v4 256-dimensional example is not a universal wire format. Future edge encoders may scale, but retrieval compatibility requires a versioned common embedding space or an explicit validated projection. Arbitrary dimension changes cannot remain invisible behind the same semantic API.

6.7 Model deployment and lifecycle management

6.7.1 Identity and artifact boundaries

The runtime separates six concepts:

1. **Family**: a scientific task with a stable typed output, such as FQE.
2. **Variant**: a trained architecture and input strategy, such as a 64-pixel mobile backbone or a multiscale larger encoder.
3. **Deployment artifact**: immutable weights, numerical format, tensor schema, preprocessing, postprocessing, calibration, and provenance.
4. **Runtime candidate**: an artifact plus execution provider, provider options, and an admitted input/batch profile.
5. **Local qualification**: measured correctness, latency, and resource evidence for that candidate on a specific runtime stack.
6. **Selected instance**: the active candidate handle and its selection epoch, operating mode, fallback, and health.

Model parameters are not retrained for every graphics card. A finite set of artifacts can be evaluated on many machines. Device-specific engine compilation and local benchmarking are separate from scientific training.

6.7.2 Manifest and trust

Manifest v2 will identify family/contract version, variant, artifact version, weights digest, member-file digests, precision, input/output schema, pre/post contracts, calibration, training lineage, frozen evaluation manifest, quality profile, lifecycle policy, and minimum runtime requirements. The bundle digest must bind preprocessing and calibration as well as weights. Promotion and revocation are separate attestations bound to that digest; they do not mutate the verified bundle. Replacing calibration or quality evidence creates a new bundle identity and invalidates affected qualification records.

Provider declarations express tested or intended compatibility; a successful local probe is still required. A hash verifies integrity against a trusted manifest, not the authenticity of a manifest downloaded alongside the weights. Any future model distribution channel needs an explicit trust root, signed manifest policy, atomic installation, revocation, and rollback rules. The first implementation may restrict itself to operator-installed local bundles.

6.7.3 Orthogonal states

Dimension	States and interpretation
Artifact lifecycle	Reference, shadow, promoted, retired, revoked. Controls scientific approval and permitted use.
Local installation	Absent, present, verified, quarantined. Describes local files and integrity.
Runtime qualification	Unbenchmarked, eligible, ineligible, stale. Applies to one candidate under one runtime fingerprint and workload profile.
Activation	Inactive, selected, fallback. Identifies runtime use; it never grants autonomy by itself.

Reference weights are test fixtures. A shadow artifact may be evaluated and logged without controlling hardware. An artifact must be promoted and locally qualified before being considered for autonomous use, and the controller must separately permit that authority. Current FQE remains shadow-only even if a bundle declares a promoted state.

6.7.4 Compute profile and execution-provider probe

Inventory CPU architecture/ISA, logical and physical cores where available, memory, accelerator identity, and power configuration. Enumerate providers present in the actual binary/runtime installation. Probe session construction, operator compatibility, memory allocation, and a golden vector.

Distinguish requested provider from actual graph placement. Partial execution on CPU or silent fallback must be observable and benchmarked as that mixed candidate. A machine with a GPU but without a compatible provider remains on an eligible CPU or classical path. CoreML and Metal are not interchangeable provider names; backend-specific implementation details belong to the release matrix.

6.7.5 Eligibility and preference

For family f, candidate c, deployment fingerprint d, and workload w, define the admissible set

$$\begin{aligned}\mathcal{E}_f(d,w) = \{c : &\text{integrity}(c) \wedge \text{contract}(c) \wedge \text{quality}_f(c) \\ &\wedge \text{permission}(c, \text{mode}) \wedge \text{parity}(c,d) \\ &\wedge L_{95}(c,d,w) \leq D_f \wedge \text{tail_policy}(c,d,w) \\ &\wedge \text{memory_reserve}(c,d,w) \wedge \text{qualified}(c,d,w)\}.\end{aligned}$$

D_f is a subsystem budget derived from the complete observing schedule. It is not a universal 50 ms limit. p95 admission alone does not guarantee every deadline; record p99, miss frequency, maximum observed latency, and downstream observation age. These are operational SLOs on a general-purpose OS, not a proof of hard real-time scheduling.

Preference uses a versioned family-specific quality policy evaluated on the same frozen split. Minimum accuracy, calibration, and critical-slice gates apply before resource preferences. Among non-dominated candidates, `efficient` favors lower resource cost, `balanced` reserves comfortable headroom, and `quality` accepts higher cost within hard limits. Use explicit tie-breakers and report the selection reason. A generic unqualified quality scalar is insufficient.

When several families share an accelerator, enforce joint memory reservations and a scheduling budget, then verify the selected combination under concurrent load. Shared/unified RAM must be counted once in the system memory envelope, not once as RAM and again as independent VRAM.

6.7.6 Benchmark method

Benchmarks use representative captured or public test fixtures whose use is permitted. FQE fixtures cover realistic crop counts, star quality, bit depth, and input shapes. Keep correctness fixtures separate from performance fixtures where necessary, and keep model-quality test labels out of runtime tuning.

Measure preprocessing, CPU-to-device transfer, queue wait, session execution, device-to-CPU transfer, aggregation, and publication. Report both component timings and complete result age. Record session/engine creation separately; exclude warmup from steady-state percentiles but retain it as startup cost. Run candidates sequentially with memory released or accounted for between trials. State run duration, sample count, percentile estimator, thermal/power state, and workload profile; do not present a short run's p99 as an all-night guarantee.

Use cold/startup, warm idle, representative co-load, and sustained profiles. The sustained profile must be long enough to observe the target's stable operating state and documented thermal behavior. If a sensor is unavailable, record unknown rather than a passing thermal result. Background qualification must yield to active acquisition and control.

6.7.7 Benchmark cache

A cache record is keyed by bundle/artifact digest, semantic/pre/post contract, daemon build and benchmark schema, ORT/provider version and options, OS and architecture, CPU ISA and thread settings, accelerator/device identity, driver/runtime, actual provider placement, input/batch profile, power mode, and concurrency/load profile. It stores timestamp, methodology, sample count, timings, deadline misses, resource peaks, correctness result, and failure reason.

Changes in those inputs invalidate qualification. Ordinary free-memory fluctuations trigger admission checks rather than an unbounded number of cache entries. A cached result is evidence

under recorded conditions, not permission to ignore current reservations, health, or a revoked artifact. Store device engine caches separately and invalidate them by their own compatibility keys.

6.7.8 Startup and activation

1. Start classical acquisition, measurements, and control services.
2. Verify installed bundles and build a complete candidate inventory.
3. Probe compiled providers; reject corrupt or incompatible candidates.
4. Reuse valid qualification records and run missing tests when the rig is idle.
5. Apply family quality/permission gates and joint resource budgets.
6. Construct the selected session and validate its readiness before publishing an atomic handle. Retain the safe baseline throughout qualification.
7. Pin selections for the observing session and emit the full selection reason.

For the initial rollout, the safe baseline is classical operation and optional qualified shadow FQE. It must not depend on the presence of a promoted neural model. Avoid loading every candidate at once or delaying hardware availability while an accelerator builds an engine.

6.7.9 Health, degradation, and switching

Initial implementation pins the selection. Inference faults still produce immediate unavailable results and classical fallback. A later phase adds conservative dynamic demotion after repeated SLO violations, with hysteresis, minimum dwell time, and a clear event. Do not opportunistically promote a model midway through autofocus or an exposure transaction.

Validate a replacement before an atomic switch at a safe boundary. Account for the temporary memory overlap of old and new sessions; warm standby is optional when memory permits, not a universal requirement. Carry a selection epoch with each result so late outputs cannot overwrite newer observations. Pin one measurement contract across an autofocus curve or restart its comparison window after a model change, because calibrated variants can still have different residual biases.

6.7.10 Evidence and limits

MobileNetV3 and EfficientNet support capacity families; Jacob et al. support validated precision-specific deployment[1, 2, 3]. MnasNet and NetAdapt support measured latency rather than FLOP-based selection[4, 5]. MLPerf Mobile and ML-EXray motivate whole-stack measurement and deployed parity[6, 7]. Research on resource-aware adaptation supports later switching[8, 9], while SBC experiments motivate recording power configuration[10]. None establishes Achernar-specific tiers, memory budgets, or accuracy gains. Those require Achernar data and hardware validation.

7 AI Parameter Control and Closed-Loop Autonomy

7.1 Control philosophy

Retain the v4 progression: classical baseline, learned perception, logged proposal, supervised bounded actuation, then carefully evaluated adaptive policies. End-to-end reinforcement learning that directly controls equipment remains simulation-first research. Faster inference does not accelerate this promotion ladder.

7.2 Formal subsystem contract

For each controller, define observation o_t, inferred state s_t, proposed action a_t, objective, and constraints. Constraints include current ownership, travel range, exposure state, minimum settling time, fresh device feedback, measurement validity, confidence/uncertainty policy, and cumulative action budget. Log the proposal, validated action, executed action, and observed outcome separately.

7.3 Daemon responsibilities

v6 requirement; implementation pending. The supervisor will manage autonomy modes and fault downgrades; the arbiter will resolve ownership and priorities. The corresponding supervisor, arbiter, and ownership types currently exist as unwired scaffolding. The focus API does instantiate `FocusSafetyGuard`, and the focus path orchestrates its classical sweep, motion, settling, measurements, and fit. That narrower implementation does not establish system-wide ownership or safety coordination. These control responsibilities remain outside the model selector.

Priority remains emergency stop, collision/safety, mount recovery, autofocus, exposure tuning, then convenience recommendations. Ownership must be released on success, error, timeout, and cancellation. Repeated model faults can reduce autonomy but cannot relax safety limits.

7.4 Database schema extensions

The current SQL migrations remain the source of truth for rigs, focuser profiles, filter offsets, focus runs/samples/predictions/actions, drift events, and model records. Preserve failed runs and rejected samples with reasons to avoid training only on successful observations.

Future runtime tables will store benchmark records, selected-candidate epochs, and activation/fallback events. Frame measurements refer to the appropriate epoch and exact artifact identity. Persist aggregate latency windows and transitions; detailed per-inference traces may be sampled or bounded by a retention policy. Runtime profiling must not create uncontrolled I/O during capture.

7.5 Focus control API

Current implementation. Routes include `/api/v1/focus/run`, `/abort`, `/status`, `/history`, `/drift`, and `/drift/history` under the focus prefix. Low-level focuser movement has its own router. The broader v4 proposal for profile editing, calibration jobs, dataset exports, and replay endpoints remains a future contract unless implemented and exposed by the current router.

Requests must distinguish manual operation, advisory/shadow evaluation, and bounded autonomy. A response reports current state, actual position if known, sample provenance, classical

measurement, shadow prediction if present, calibration, and failure reasons. Publish state transitions and action events without claiming that an unverified hardware movement succeeded.

7.6 ML model specifications

7.6.1 Model 1: Focus Quality Estimator (FQE)

Current implementation. The architecture in `ml/models/focus_quality_cnn.py` is a grayscale MobileNetV3-Small backbone, global pooling, and a two-output head. Input is a normalized star crop $(B, 1, H, W)$; the current daemon/export profile uses $H = W = 64$. Outputs are `mu_log_hfr` and `log_var`. Focuser position is excluded from image-only FQE to avoid rig-specific shortcuts; it belongs in FOP features.

For target $y_i = \log(\mathrm{HFR}_{i,\mathrm{px}})$, predicted mean μ_i, and log variance s_i, the current training objective includes Gaussian NLL:

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{N} \sum_i \left[\frac{1}{2} \exp(-s_i)(y_i - \mu_i)^2 + \frac{1}{2} s_i \right].$$

The training code clamps log variance for numerical stability and supports an auxiliary within-run ranking loss. Per-star uncertainty and post-pooling frame uncertainty are calibrated on held-out groups and bound to the artifact.

The current robust frame estimate is the exponential of median predicted log-HFR. Its uncertainty combines robust between-star dispersion and mean aleatoric variance, followed by a separately fitted frame scale. This formula is an implemented aggregation rule, not proof that stars are statistically independent or that nominal interval coverage is correct. Coverage must be measured at frame and run level.

Proposed variant	Possible compute expenditure	Required evidence
Nano	Mobile backbone; 64-pixel profile; FP32 or validated INT8 artifact	Minimum quality and CPU deployment baseline.
Medium	Wider backbone; larger/native crop or increased star count	Better task quality within measured latency and memory.
Large	Larger CNN or validated hybrid; 128-pixel or multiscale profile	Improvement on difficult observing slices, not parameter count alone.
Ensemble	Several validated members or repeated stochastic inference	Incremental quality and uncertainty benefit including full latency cost.

These are research candidates, not finalized or shipped architectures. Larger native crops and resizing a fixed field of view are different transformations. If resizing changes the scale, postprocessing must restore HFR to original sensor pixels. Crop selection, edge exclusion, maximum star count, batching, padding, and aggregation are versioned and included in parity and quality tests.

The current classical source gates determine which stars reach FQE. A neural model cannot improve rejected cases it never sees. Evaluate both matched accepted-star cohorts and full frame coverage; redesigning source selection is a separate validated change. Agreement with classical

HFR labels alone does not establish superiority to the classical baseline. Use synthetic PSFs with known properties, independent fitted measurements where available, and operational focus outcomes to investigate additional benefit.

7.6.2 Model 2: Focus Offset Predictor (FOP)

FOP predicts signed focuser steps and uncertainty from quality, position, temperature, filter, history, direction, rig, and sky features. The current canonical 38-feature contract is `ml/contracts/fop_features.py`. Do not reproduce inconsistent hand-counted feature dimensions in exporters. Reliable absolute position and successful run optima are prerequisites for training labels.

Keep a single validated CPU implementation, initially a tree model if that meets measured requirements. The v4 claim that LightGBM necessarily requires MLP distillation for fast inference is withdrawn: distillation needs an actual deployment bottleneck and a new quality evaluation. Optional future FQE embedding fusion requires a versioned feature contract; changing FQE variants cannot silently substitute incompatible hidden embeddings.

7.6.3 Model 3: Focus Drift Forecaster (FDF)

The current state is $x = [p, v, k]^T$: best-focus position, thermal-independent drift rate in steps/s, and temperature coefficient in steps/°C. For measured temperature change ΔT and elapsed seconds Δt:

$$F = \begin{bmatrix} 1 & \Delta t & \Delta T \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad p' = p + v\Delta t + k\Delta T.$$

The measured exogenous temperature regressor keeps the model linear and time-varying. The implementation uses process noise for position/velocity, a slow coefficient random walk, and a Joseph-form covariance update. This supersedes v4's two-state example and separate exponential coefficient update.

Estimation and refocus advice are separate: missing optics or critical-focus threshold can make advice unavailable while identification continues. Update from verified autofocus completion after the hardware move succeeds. Persist mutable state in the data directory. There is no drift ONNX artifact.

7.6.4 Model 4: Cloud / Obstruction Classifier (COC)

Begin with a small current-condition model and a defined sky/input mask. Public region annotations motivate an optional segmentation auxiliary task, not an automatic five-class conversion. Preserve cloud presence/coverage, transparency or attenuation, and contamination as distinct proposed quantities. The existing five diagnostic names remain legacy annotations; freeze their operational meaning, including thin cloud, before any binary collapse. Contract changes require versioned model, exporter and daemon work. Temperature scaling applies to classification probabilities; it does not automatically calibrate regression uncertainty[11].

Evaluate high-recall cloud detection, false pauses, missed thin cloud, calibration, and site/season/moon/optics slices. Keep temporal hysteresis outside the selected network so a model replacement cannot reset or bypass pause policy. A cloud image classifier cannot certify the absence of rain or replace weather sensors.

7.6.5 Model 5: Trailing / Elongation Detector (TED)

The baseline measures source elongation, severity, and orientation with explicit source-quality gates. Future ambiguous-case crops may use a small CNN. Preserve classical output during neural failure. A higher-capacity variant must show reduced confusion between tracking error, defocus, optical asymmetry, wind, and blending on labeled data. Evaluate whole-frame operational false alerts as well as per-crop classification.

7.6.6 Model 6: Background / SNR Estimator (BGE)

Classical statistics remain the required path. SEP-style background estimation and the current implementation must be distinguished in status documentation; describing SEP in the architecture does not mean the daemon already uses it. The optional neural noise-map estimator is future work with separate validation of gain, noise, bias, and downstream photometric effects. GPU presence alone does not activate it.

7.6.7 Model 7: Exposure / Gain Recommender

The proposed contextual bandit uses a small feature vector and bounded action space. For arm a, maintain A_a and b_a, with

$$\hat{\theta}_a = A_a^{-1} b_a, \qquad u_a(x) = \hat{\theta}_a^T x + \alpha \sqrt{x^T A_a^{-1} x}.$$

Numerical implementations should solve linear systems rather than form an inverse unnecessarily. Update from attributable outcome rewards, including acceptance, SNR, saturation, and trailing, with versioned weights. Exploration is limited to safe actions and can be disabled during unstable conditions. The algorithm remains CPU-side across hardware tiers.

7.6.8 Model 8: Filter Order Recommender (FOR)

A future tabular recommender combines target requirements, sky conditions, remaining integration, and prior outcomes. Maintain one validated CPU artifact. Recommendations pass through sequence ownership, filter-focus coordination, and exposure boundaries. Feature explanations aid diagnosis but do not prove causal effects or authorize motion.

7.6.9 Model 9: Embeddings and future AstroM3Edge

Retain cloud multimodal retrieval, ranking, and anomaly research. Optional image, light-curve, spectrum, and metadata encoders require explicit alignment training and missing-modality behavior. A future local image/telemetry encoder can join the compute-aware registry only after its semantic space, output contract, validation data, and runtime budgets are defined.

7.6.10 Model filesystem layout and metadata

Current implementation. Bundles use `models/<model_name>/<version>/`. The FQE directory contains ONNX, metadata, normalization, and calibration files. The current reference bundle must remain marked ineligible.

Proposed organization:

```
models/<family>/<variant>/<artifact-version>/
  manifest.json       # schema v2; member digests and contracts
  model.onnx
  preprocess.json
  postprocess.json
  calibration.json
  quality.json        # frozen evaluation identity and task metrics
  golden/             # reference inputs, outputs, tolerances

<data_dir>/model-runtime/
  benchmarks/         # local qualification; may instead use SQLite
  engines/            # provider/device-specific compiled caches
```

Names are a proposed layout, not files created by this documentation revision. Migration must inventory all older versions before choosing candidates. A newer ineligible artifact must not hide an older eligible artifact. A legacy adapter may admit the existing shadow contract after verification, without inventing missing quality or promotion evidence.

7.7 Training pipeline specification

7.7.1 Dataset contracts

FQE examples identify frame asset, crop coordinates, run, date/session, rig, sensor/optics configuration, label method, validity, and physical units. FOP examples add position, direction history, best-fit run optimum, uncertainty, and filter offsets. Dataset schemas and feature order are executable contracts under `ml/contracts` and dataset builders, not informal prose alone.

7.7.2 Split strategy

Group samples by observing run and session; adjacent crops/frames cannot cross train and test. Freeze chronological splits and hold out rigs or optical trains for generalization. Keep calibration data separate from final test data. Every compared variant must use the same manifest, allowed population, and slice definitions. Report exclusions and coverage, including failed runs.

7.7.3 Preprocessing

Training and daemon preprocessing must match numerically within documented tolerances: channel interpretation, bit depth, black/saturation levels, crop coordinates, clipping/interpolation, normalization, and resizing. Bind pre/post files to the bundle. Test Python and Rust against the same golden fixtures, including empty, saturated, edge-truncated, and invalid inputs.

7.7.4 Training stages

Extract and validate data; freeze groups; train independent variants; calibrate; evaluate all slices; export a deployment artifact; verify numerical parity; measure the deployed pipeline; publish as shadow; evaluate live shadow behavior; then consider promotion. Synthetic-only and random artifacts remain test-only. Keep independent variants for the first rollout. Elastic supernetworks such as Once-for-All are a later training strategy[13].

7.7.5 Project layout

The existing ML modules cover data preparation (`ml/data_pipeline`), preprocessing (`ml/preprocess`), model definitions (`ml/models`), training (`ml/train`), evaluation (`ml/eval`), export (`ml/export`), and orchestration (`ml/pipelines`). Exporters will generate v2 manifests, golden vectors, and quality reports. Do not download or train models merely to satisfy a runtime unit test; hermetic reference artifacts exercise contracts without quality claims.

7.8 Focus controller algorithm

Current implementation. The operating focus path uses a classical sweep and fit with fresh frame capture, action logging, and completion/error handling for the move to the fitted optimum. It does not yet acquire a fresh optical verification sample after that final move; the fitted HFR is not a post-move measurement. FQE predictions are shadow measurements. The bounded AI offset loop remains subject to training, replay, and control-promotion work.

The future loop captures a baseline, validates the observation, obtains an offset proposal, checks uncertainty and ownership, clamps travel, moves and settles, measures a fresh frame, and tests improvement. It reduces step size or falls back to a classical bracket/curve fit when improvement is absent. Total travel, duration, failed-improvement count, and uncertainty cap the run. Do not reuse stale inference after motion, filter change, stream restart, or a model-selection epoch change. The completed bounded controller must acquire a fresh optical sample after moving to its chosen optimum before declaring verified focus quality; successful motor-command completion alone is insufficient.

7.9 Focus configuration

Rig-specific travel and backlash values must be measured. The illustrative numbers in v4 are not safe universal defaults. Preserve the current `[focus.safety]`, `[focus.backlash]`, `[focus.models]`, and `[focus.drift]` parser contracts until a code migration changes them. Documentation examples of future AI settings must remain commented or explicitly labeled so they cannot be mistaken for active controls.

7.10 Evaluation and promotion criteria

7.10.1 Model metrics

FQE: HFR error in pixels, within-run ranking, interval coverage/width, frame/run bias, and coverage of difficult stars. COC: precision/recall, false-pause rate, missed clouds, and calibration. TED: precision/recall, severity error, and false guiding diagnoses. BGE: background/noise estimation error and downstream scientific bias. FOP: step error, direction errors, uncertainty coverage, and operational move count.

Aggregate metrics must not hide failures on low-SNR, saturated, crowded, poor-seeing, unusual-optics, or held-out-rig slices[12]. Report the sample count and uncertainty of comparisons. A larger model is preferred only when the relevant quality policy demonstrates benefit.

7.10.2 Operational control metrics

Measure focus convergence, final classical HFR, elapsed focus time, movement count, fallback/clamp/abort rates, false pauses, retained exposure quality, and operator overrides. Evaluate

each candidate's complete deployed behavior. Numerical speed alone does not justify promotion.

7.10.3 Testing stages

Progress through offline validation, deployment parity, historical replay, live shadow evaluation, controlled lab actuation, supervised field use, and then any broader rollout. A promotion record identifies the exact bundle, quality report, calibration, data split, approved use, and rollback target. Runtime qualification must be repeated when the deployment fingerprint changes.

7.11 Autonomy maturity ladder

Retain L0 classical baseline, L1 perception/advice, L2 logged proposal, L3 supervised bounded autonomy, L4 validated per-rig adaptation, and L5 coordinated observatory operation as a long-term objective. Implementation milestones are earned per subsystem. A capable GPU never raises this level.

8 Filter Wheel, Mount, and Shared Perception

8.1 Shared perception bus

Controllers consume typed `SharedObservation` values rather than raw pixels or provider-specific tensors. The vision layer owns source extraction, image preprocessing, model execution, and aggregation. The intended cadence is 1–5 Hz; current startup calls the vision loop at 1 Hz.

Preserve units and missingness: HFR and HFD are in pixels; HFD equals twice HFR, but FWHM is not generally twice HFR. Angular image FWHM requires an actual fit and pixel scale, and is not automatically an atmospheric seeing measurement. The code has removed arbitrary reciprocal sharpness and composite frame scores.

v6 requirement; implementation pending. Add frame identity, per-family validity/freshness, and selection provenance without changing scientific semantics. Missing neural output must not invalidate an otherwise valid classical measurement. Missing weather or mount telemetry must remain missing. Plate-solve state has an independent timestamp and cannot inherit a fresh timestamp from a new vision frame.

Measurement	Permitted downstream use
HFR and uncertainty	Focus fit/verification and quality monitoring. Preserve classical and shadow sources separately.
Cloud evidence	Pause/recovery policy with hysteresis and freshness; corroborate weather safety with the appropriate independent sensors.
Trailing evidence	Exposure quality and advisory guiding diagnosis; require coordinate/telemetry evidence before attributing a mount axis.
Background and SNR	Exposure/gain context and filter comparisons using calibrated units and comparable exposures.
Plate-solve result	Pointing verification and correction with its own age, uncertainty, and geometry constraints.

The runtime uses bounded workers and backpressure. Processing all models in parallel is

permitted only after joint resource qualification. Drop obsolete preview work deliberately, while preserving required autofocus samples and acquisition provenance.

8.2 Filter wheel subsystem

8.2.1 Filter-focus handshake

At a safe exposure boundary, acquire wheel and focuser ownership, verify the current and target filters, rotate, verify completion, settle, apply the relative offset between the two filter references, then verify focus when the offset is uncertain. Record the transition and release ownership on all paths. Unknown position or an incomplete handshake blocks automatic exposure resumption.

8.2.2 FW-1: Classical per-filter sky scorer

Estimate signal/noise in consistent electron units. An illustrative aperture model is

$$\mathrm{SNR}(f,t) = \frac{S_f t}{\sqrt{S_f t + n_{\mathrm{pix}}(B_f t + D t + R^2)}},$$

where S_f is source electrons/s in the aperture, B_f and D are per-pixel background and dark rates, and R is per-pixel read noise in electrons. Real comparisons require throughput, gain, exposure, and source assumptions. Do not mix ADU rates with electron variance or treat this simplified model as a calibrated estimate without those inputs.

8.2.3 FW-2: Contextual filter sequencer

The proposed small CPU bandit combines sky evidence, remaining integration, switch cost, target requirements, and recent outcomes. An explicit continue arm avoids unnecessary mechanical changes. Freeze exploration when observations are invalid or weather/recovery policy owns the sequence. Narrowband performance under moonlight and cloud attenuation must be evaluated separately; clouds do not become safe or harmless merely because a narrowband filter is selected.

8.2.4 FW-3: Session plan optimizer

The proposed CP-SAT scheduler assigns discrete integration slots to filters subject to visibility, total time, minimum integration, switching, and focus overheads. CP-SAT variables and coefficients must use a documented integer scaling. Linearize products of choice and integration variables or use fixed slots; the continuous bilinear objective in the older sketch is not directly an executable CP-SAT formulation. Solve time is measured on target hardware; the prior universal two-second assertion is a target requiring evidence.

8.2.5 Filter safety configuration

Enforce no switching during exposure, minimum settling, verified wheel position, switch-frequency limits, and focus coordination. Model selection must not override a locked observing plan or these requirements.

8.3 Mount subsystem

8.3.1 M-1: Guiding aggressiveness tuner

A future pair of CPU bandits may tune bounded RA/Dec aggressiveness using actual guider telemetry and quality feedback. Missing guider data prevents learning. Seeing, wind, clouds, and mount behavior can confound reward; freeze or reject updates when cause and outcome cannot be attributed adequately.

8.3.2 M-2: Dither policy

A small fixed CPU regressor may recommend magnitude and settle threshold. Units, image scale, feature ordering, and true input dimension require an executable contract; the v4 hand-counted dimension conflicts with its listed one-hot features and must not be copied into a new model. Dithers occur only between exposures with appropriate settle verification.

8.3.3 M-3: Plate-solve trigger

A small tabular classifier may recommend when to solve. Star-count changes can arise from clouds, focus, exposure, or pointing and are not alone proof of pointing drift. Deterministic startup, post-slew, post-flip, and recovery rules remain authoritative until the learned trigger is validated.

8.3.4 M-4: Pointing model

A regularized spatial correction can model systematic pointing errors from successful solve residuals. Use a coordinate treatment that handles RA wrap and polar geometry, and retain coordinate epoch/frame metadata. Limit extrapolation and correction magnitude, validate on held-out sky regions, and recheck mount geometry before motion. A raw planar spline on unwrapped RA/Dec is only an illustrative historical sketch.

8.3.5 Mount safety

Horizon, altitude, pier geometry, meridian policy, motion limits, ownership, park verification, and weather interlocks remain independent of perception capacity. No automatic GPU policy adds permission to slew. Every broader autonomy release requires real hardware verification of its complete safety path.

8.4 Filter and mount schemas

Existing migrations define filter switches, guider logs, plate-solve events, and pointing snapshots. Future records must retain request/outcome identity, measurement age, policy version, and runtime provenance when a learned estimate contributes. The presence of a database table does not establish a live subsystem.

8.5 Subsystem compute classes

Family or algorithm	Compute strategy	Priority/status
FQE	Scalable perception	First runtime pilot; real data and promotion needed.
COC	Scalable perception	Second, after corpus and executable model.
TED neural branch	Optional scalable perception	Later; classical first.
BGE neural branch	Optional scalable perception	Later; scientific validation gate.
AstroM3Edge	Future scalable encoder	Research; stable embedding contract needed.
FOP and FOR	One validated CPU artifact	Data/control prerequisites govern readiness.
FDF	Fixed CPU estimator	Current three-state filter; mutable rig state.
Exposure/gain, FW-1/2/3	CPU rules, bandit, optimizer	No GPU tier requirement.
M-1/2/3/4	CPU bandit, small regressor, trees, spatial model	No GPU tier requirement.

9 Data Engineering and Labeling

9.1 Classical labels and measurement lineage

Use classical HFR, source masks, background, and elongation as documented labels or features, retaining raw and derived asset references. Label precision and measurement failures set a ceiling on what supervised training can infer. Include failed autofocus runs and invalid conditions with explicit labels or exclusion reasons.

9.2 Weak labels

Rules for cloudy, saturated, trailed, or accepted frames are useful weak supervision. Store the rule version and disagreement with human review. Acceptance decisions produced by the incumbent model require safeguards against a feedback loop that excludes its own failure cases.

9.3 Public archives

Use public archives first for generic perception and native catalog research. Every source needs a versioned asset, label, group, processing and intended-use manifest before ingestion. Noncommercial or unresolved sources remain in a separately tracked research lineage. Source labels and raw or processed pixels are not interchangeable. The next section defines the revised evidence gates. Survey success is not evidence of R50 or amateur-optics performance.

9.4 Augmentation

Apply physically motivated perturbations with known label effects. Blur can change the HFR target; it cannot be applied while retaining the original focus label without justification. Simulate noise, background gradients, hot pixels, missing observations, and resolution changes within the task contract. Evaluate synthetic-to-real gaps explicitly.

10 Public-Data-First Development and Minimum Local Evidence

10.1 Revision decision and authority

The October 12 audit materially changes the collection strategy. Public astronomical imagery and controlled simulations can supply broad perception examples; actual instrument calibration and operational outcomes remain local. This is a change in data priorities, not a replacement for the local daemon, classical measurement authority, bounded controller or v5 runtime gates.

The source ledger and bounded file-inspection evidence are in `docs/public-dataset-audit-2026-10-12/README.md` and its three specialist memos. The current acquisition plan is `achernar_data_collection_v6.tex`. The original `achernar_data_collection.tex` is retained as a dated historical campaign/procedure source; its quotas and old defects do not override v6. Research sources were checked on October 12, 2026. No model was trained, no transfer accuracy was measured and no local hardware was actuated by that audit.

10.2 Dataset suitability has several independent gates

For each use, establish access, permitted intended use, suitable pixels and labels, recoverable grouping, preprocessing compatibility and task value. A passing access check is not a passing label or deployment check. A public code license does not automatically license data, teacher weights or model checkpoints. Restricted sources and their descendants remain identifiable; the product route requires documented permission for its intended uses.

Resource	Verified useful role	Limitation / initial decision
LenghuSky-8	Nighttime cloud human regions, processed images, teacher logits	Start with 1,111 image-bearing JSONs (22.1 MB); bulk logits are weak labels.
SWINSEG / SWINySEG	Small Singapore cloud benchmark / augmentations	Shared originals forbid independent train/test use; CC BY-NC 4.0.
Almeria v2	818 daylight image/mask pairs, including 48 test pairs	CC BY 4.0; cloud-layer labels and geometry differ from local night conditions.
DeepSky / WOANC	Optional extra classification / nighttime domains	DeepSky daytime archive unresolved; WOANC author-request data, not a blocker.
MILAN / DeepSpaceYolo	Raw small-telescope FITS / stacked object detection	CC BY-NC-ND; March MILAN package is 49.4 GB, not the full survey.

Resource	Verified useful role	Limitation / initial decision
HSC PDR3	Single-visit processed exposures and fitted PSFs	Registration/noncommercial terms; do not describe coadds or calexp as raw.
MAST	Selected public mission images, masks and uncertainty products	Approve actual mission/product/units and terms; archive umbrella is not a dataset.
GalSim	Controlled optical/atmospheric/noise image formation	Software and generator parameters; not measured rig data or motor steps.
BTSbot / Trailblazer	Optional transient/residual/satellite challenges	Not primary star-tracking labels; per-file rights/access need checking.
LCO	Calibration/shape experiments and focus-related header histories	CC BY-NC 2.0; sampled verification frames do not establish full focus sweeps.
Rubin OpSim / PHD2	Scheduling scenarios / guiding parser and replay	Different objective or uploaded-log rights; missing policy counterfactuals.
AstroM3	Native variable-star photometry/spectrum/metadata inference	21,440 objects in a full configuration; IDs absent from processed vectors.
Multimodal Universe	Select approved image subsets and candidate matches	Subset terms and object identity; no 100 TB download or universal edge alignment.

These are bounded research judgments, not quality results. Exact release URLs, license distinctions, sizes, hashes, sampled shapes and unresolved access are recorded in the specialist memos. The same source can be useful for an offline research benchmark and ineligible for a different distribution purpose.

10.3 Highest-impact corrections

Lenghu. Its 429,620-frame release is mostly teacher-labeled; the 1,111 human images are the complete partially annotated benchmark, not an untouched evaluation set. Unannotated pixels are ignore, not clear sky. The small ZIP embeds all human-benchmark inputs, so initial experiments need not transfer the roughly 79 GB bulk release. Reviewed label spelling/type defects require a correction ledger. Exclude holdout ancestors and adjacent same-night sequences from teacher-training expansion.[14, 15]

SWIN. SWINySEG contains transformed SWINSEG originals; its 6,768 files are largely daytime and include augmentation. Using the former to train and the latter to test leaks. Hold out the whole source family when assessing independent transfer.[16]

AstroM3. One full configuration has 21,440 examples; the headline 5.37 GB sums overlapping seed/subsample packages. Its native modalities are photometry, spectrum and astrophysical metadata. An image adapter also needs source identity and a defined alignment objective. Guider telemetry is not photometry and capture settings are not the native astrophysical feature vector.

Reused weights are initialization, not demonstrated alignment.[17]

Rights. MILAN/DeepSpaceYolo are CC BY-NC-ND; HSC archive terms are noncommercial scientific/educational; LCO states CC BY-NC 2.0. Those restrictions are not removed by using an open-source simulator. No blanket legal conclusion about whether every trained weight is an adaptation is made here; preserve research lineage and document coverage before distributing a candidate.[18, 19, 20]

10.4 Source, label and split manifests (proposed)

An archive adapter shall preserve original assets and emit a dataset manifest separate from daemon operational tables. Do not create fictitious autofocus runs, positions or temperatures merely to satisfy the current builder. Required fields are:

- Provider, exact release/revision and source URLs; asset ID and checksum; separate terms for code, data, checkpoints and teachers; intended-use disposition.
- Acquisition system, native geometry/scale, units, CFA/channel layout, calibration level, masks, black/white/clipping evidence, timestamp and clock quality, site/night/session and camera-era identity where available.
- Parent exposure/object/scene IDs; transformation and simulation versions, seeds, crop origin, resize geometry, preprocessing/calibration hashes.
- Label type (human partial region, measured reference, fitted product, teacher prediction, latent generator parameter, noisy estimated label), method, uncertainty/validity, correction history and exclusions.
- Frozen split assignment, ancestor group and rationale; source membership; separate adaptation, calibration and final-test roles; dataset revision for every run.

This contract is not yet implemented by adding prose. Current FQE ingestion joins Achernar SQLite focus records and preserved grayscale assets. FITS/NPY adapters, masks/variance handling and source-family split validation remain work.

Group before augmentation. All derivatives of an exposure and all views of a matched object stay together. Nested run/night/rig or camera-era constraints must match the evaluated claim. Temporal grouped splitting already exists in `ml/data_pipeline/split_registry.py`; random run splits do not by themselves prove independent-night or unseen-rig performance. Unknown dates require conservative session families, not fabricated UTC. Calibration and model selection use different groups from the untouched final test. Selecting new training data from final-test failures consumes that test; retire it and obtain new independent test evidence before another final claim.

10.5 Measurement and simulation semantics

FQE reference HFR is not instrumental defocus or a signed motor offset. Seeing, motion, sampling, blending and background affect width. Agreement with a single classical labeler establishes imitation of that reference, not physical superiority or better autofocus. Preserve independent fitted/analytic checks and post-move outcomes. Archive-only stars do not support real within-run focus ranking; disable that objective or use explicitly tagged synthetic sequences.

Keep latent defocus/motion/noise parameters separate from the noise-free reference after pixel integration and from noisy estimator outputs. Do not convolve a noisy real crop and call it an independent physical exposure. Model noise correlations and original motion honestly. Record pupil obstruction, PSF wings, subpixel phase, aberrations, CFA response, clipping and rejected

sources. A fixed straight trail cannot represent all jitter/periodic error. Motion angle is axial (modulo 180 degrees); an isotropic source has no useful orientation target. WCS is required before sensor axes become sky axes.

As an illustrative derivation, a circular Gaussian of width σ convolved with a uniform line of length L has ideal moment ellipticity

$$e = 1 - \frac{\sigma}{\sqrt{\sigma^2 + L^2/12}}.$$

Thus exact latent trail length does not supply a universal observed elongation label. The old sidereal-drift formula predicts ideal displacement under stated geometry, not exact thresholded moments or the cause of a distorted star.

10.6 The local evidence that remains necessary

Family	Public/synthetic work first	Local work only for the supported domain
FQE	Reference star widths, noise/PSF stress tests, one small candidate	Mode/preprocessing parity and real independently verified focus outcomes.
FOP	Feature/curve simulation and robust classical fitting	Position/sign/scale, reversal/backlash, validated optimum and final-move outcomes.
FDF	Existing three-state estimator and simulated identifiability tests	Passive ambient-temperature and verified focus histories; thermal/time variation.
COC	Human Lenghu pilot before pseudo-label expansion	R50 clear/thin transfer, glow/dew/foreground, untouched independent episodes.
TED	Classical moments and known latent motion/aberration stress tests	Compact ambiguous-range ladder, round cases and real confounders.
BGE	Classical benchmarks and approved calibrated public products	Actual supported-mode gain, read noise, linearity/clipping and relevant dark behavior.
Exposure / FOR	Deterministic objective and physics/scheduling scenarios	Executed actions, safe support, rewards/overhead, probabilities when randomized.
Native TME / Edge	Native inference; selective matched-image research	No dedicated local campaign; Edge waits for an identifiable useful alignment task.

Only the actual enabled rig/modes need initial calibration. Do not calibrate five possible bodies or acquire a filter wheel merely to populate speculative features. Cross-rig qualification is required before a cross-rig claim, not before a one-configuration prototype.

The existing cloud audit records 533 readable R50 originals with unreliable clock, strongly correlated bursts and provisional annotations. Reuse/adjudicate those first under the final

transform. No reliable whole-frame clear/thin set was established. Preserve invalid/unknowns as challenge cases and do not turn nearby weather or absent stars into per-frame truth.

The former FQE 60/150-run, COC class-thousands and TED 3,600-frame quotas cease to be training prerequisites. They are historical possible expansion scales, not necessary or sufficient universal validation targets. Initial tranche caps in the collection plan bound effort only. Expand for a named residual, missing condition or confidence limit; stop neural work if the classical baseline already satisfies the need without demonstrated incremental value.

10.7 Control-policy evidence and rig identification

Historical observations lack outcomes of unchosen actions. Unbiased policy comparison requires valid logging/causal assumptions and action support, not just large image counts. Record context before the decision, allowed/safe arms, chosen/requested/executed action, actual setting, policy/version and known probabilities when randomized, measurement ages, frame lineage and attributable reward components. Overrides and interrupted outcomes remain explicit. Unchosen rewards are missing, never zero.[21]

For FDF, fit/update the existing $[p, v, k]$ state rather than copy a foreign $\Delta p/\Delta T$ ratio. Near-linear cooling in time can confound v and k. Independent thermal trajectories and held-out forecast residuals matter more than thousands of repeated readings. Filter changes, position resets and successful final moves must be recorded. Public LCO offsets are in multiple physical coordinate conventions, not Achernar integer steps.

FW-1 needs filter response calibration; FW-2 and M-1 need attributable policy outcomes; FW-3 needs explicit constraints, not learned weights. M-2 requires dither/settle/science outcomes, M-3 solving/recovery labels, and M-4 this mount's solved pointing residuals. Harvest during ordinary operation once supported hardware exists. Public PHD2 logs remain parser/replay inputs unless permissions and decision evidence support another use.

10.8 Evaluation and evidence-governed development

Freeze the supported domain, baseline, metric, failure event and acceptable uncertainty before expanding acquisition. Report bias/coverage by SNR, optics, moon/sky, camera mode and group. Confidence intervals must respect correlated nights/episodes/runs; report group counts alongside crops/frames/pixels.

For illustration only, zero failures in n independent stationary Bernoulli trials yields a one-sided 95% upper bound $1 - 0.05^{1/n}$: at least 59 such trials are needed for an upper bound below 5%. Adjacent frames are not independent trials. This calculation explains why pilot completion is not promotion.[22]

Development proceeds by evidence: freeze contracts and source manifests; inspect small payloads; build classical benchmarks; test small public-trained candidates; acquire only the local calibration and independent examples needed for measured gaps; check deployment parity; then evaluate live shadow behavior. Native AstroM3 research is independent. Additional rigs, seasons, learned control policies and Edge alignment follow demonstrated need and valid evidence, not a calendar deadline or a required count of trained models.

The follow-up AI synthesis's perception / rig modeling / planning separation is useful explanatory structure and compatible with v5. Its lifecycle must not collapse artifact approval, installation, runtime qualification and controller authority. Preserve those orthogonal states and the current shadow-only FQE.

11 Metrics

11.1 Local perception quality and runtime

Maintain independent quality and runtime reports. Quality reports use common frozen data, calibration and slice gates. Runtime reports use the exact deployment fingerprint and workload. Store reproducible methodology, sample counts, confidence/coverage evidence, and the decision policy version.

At the 1–5 Hz design cadence, one interval is 1000–200 ms. This interval includes more than ONNX execution. Derive per-family deadlines from acquisition, classical processing, queueing, inference, publication, and safety headroom; no arbitrary percentage is certified here. Monitor observation age and missed deadlines through the entire path.

11.2 Recommendation metrics

Evaluate exposure and filter policies against a documented baseline using appropriate logged propensities and off-policy methods where applicable. Report uncertainty and support limitations; a biased historical log cannot prove a new policy's safety or superiority.

11.3 System-level outcomes

Track focus convergence and duration, final image quality, retained integration, false interruptions, fallback frequency, recovery success, and user overrides. Resource telemetry includes latency distributions, memory peaks, thermal state where available, and contention. Telemetry explains performance without becoming an unbounded recording workload.

12 Training and Serving Stack

12.1 Development infrastructure

Python/PyTorch and existing dataset/export/evaluation tools produce artifacts. Parquet and reproducible manifests support lineage; experiment tracking, DVC, and richer feature stores remain choices whose implementation status must be documented. Independent FQE variants provide the simplest initial comparison.

12.2 Serving infrastructure

Current implementation. Rust ORT dependency uses `download-binaries`; the FQE session does not register accelerator providers. A future release matrix must specify which providers are compiled/packaged, native library prerequisites, tested operator/precision combinations, and device validation.

CPU remains the initial baseline. NVIDIA Linux is a first accelerator candidate; Jetson aarch64 requires its own compatible build and testing. Windows DirectML, Apple CoreML, and Intel/OpenVINO are later candidates. Original Jetson Nano and Jetson Orin Nano are different devices and must never share a support assertion based on the word "Nano." Exact support is determined by the release matrix, not by this architectural list.

13 Local and Cloud Responsibilities

13.1 Local operation

Acquisition, physical measurements, qualified inference, arbitration, safety, and offline logging remain local. A high-capacity workstation model can still be a local model. “Local” describes placement and availability, not a fixed maximum parameter count or a universal precision.

13.2 Cloud operation

Training, archive-scale analysis, optional retrieval, model distribution, and coordination can use cloud resources. Network failure must not interrupt the required local baseline. Cloud advice or downloaded models do not bypass local authorization, promotion, runtime qualification, or equipment constraints.

14 Network Layer

Retain local connectivity, optional VPN/remote access, and the daemon-to-cloud relay design. Claims of WiFi access-point setup, WebRTC/TURN, or specific latency must reflect actual deployed components. Bound reconnect retries, detect stale remote state, and distinguish accepted command from confirmed hardware completion. Model downloads and uploads should be scheduled so they do not interfere with acquisition or storage throughput.

15 Web Platform and Operator Experience

15.1 Control and data planes

The current React client and Tauri shell share an operator interface; Django supports remote transport where wired. The daemon is the authority for live equipment capability and status. Local HTTP, telemetry, and relay payloads must agree on missingness, command identity, and error meanings.

15.2 Runtime status experience

v6 requirement; implementation pending. The UI shows selected family/variant, operating mode, reason, baseline, and health. Normal users choose efficient, balanced, or quality; advanced diagnostics can reveal provider placement, memory, benchmark age, and rejection reasons. Explain why a larger model is unavailable without requiring users to interpret CUDA operator errors. An unavailable model is not a failure of the whole observing system when the baseline remains valid.

15.3 Cloud ML services

Embedding/retrieval, ranking, anomaly analysis, and feedback training retain separate release and validation status. Future distribution of artifacts must publish identity and approval evidence. Preserve privacy preferences and do not upload science frames merely to benchmark a local model.

16 Pragmatic Requirements

1. Preserve raw assets and measurement lineage.
2. Start with a working classical baseline and honest unavailable values.
3. Spend extra compute only where measured quality improves.
4. Qualify exact artifacts and deployed pipelines on the target device.
5. Keep promotion, local performance qualification, and control authority separate.
6. Bind normalization, units, aggregation, calibration, and weights together.
7. Protect camera/control headroom and validate concurrent workloads.
8. Retain grouped data splits, difficult slices, and failed-run evidence.
9. Make fallback, provider placement, and selection reasons observable.
10. Preserve rollback and require live shadow evidence before actuation.

17 Implementation Roadmap and Acceptance Gates

17.1 Phase 0: Contracts and documentation

V6 adds a public-source audit, clean revised collection plan, source/label/split requirements and a bounded one-configuration research milestone. V5 established the runtime direction and detailed ADR. This documentation does not implement archive adapters, provider loading, training, or production model selection.

17.2 Phase 1: FQE runtime foundation

Implement manifest parsing and legacy shadow migration, complete candidate inventory, CPU provider probing, golden parity, benchmark/cache records, family handles, and read-only status. Keep classical authority and existing batched inference. Test reference artifacts as fixtures without claiming quality improvement. Acceptance: unavailable/corrupt models cannot prevent baseline operation, stale caches cannot qualify a changed candidate, and all selection/rejection reasons are inspectable.

17.3 Phase 2: Scientific variant evidence

Begin with one small FQE candidate against direct classical measurement and independent reference/outcome checks. Use common source/local partitions and compare synthetic, approved archive-real and mixed training without changing architecture simultaneously. Train an additional capacity variant only after the first experiment identifies useful value or a residual worth addressing. Calibrate and compare quality, coverage, hard slices and deployed parity. Acceptance is a demonstrated task benefit within the budget, not completion of a multi-model inventory; promotion remains a separate evidence-backed decision. Scientific pilots can use the existing offline tools and shadow path without waiting for a generic runtime redesign. Detailed comparisons and tranche decisions are in `docs/public-dataset-audit-2026-10-12/experiment-design.md`.

17.4 Phase 3: Accelerator release qualification

Package a tested NVIDIA Linux path and separately qualify Jetson aarch64. Exercise cold setup, engine cache, sustained and co-load performance, resource exhaustion, provider failure, and actual graph placement. Acceptance: release records identify exact device/runtime combinations

and preserve baseline recovery under failure. Other GPU platforms follow only with equivalent evidence.

17.5 Phase 4: Additional families

COC can begin its small public-human-label experiment in parallel with FQE scientific work, after its input and label contract is frozen. Runtime-family integration follows an executable, evaluated model. Extend shared budgets and concurrent qualification. TED and neural BGE follow demonstrated task value and scientific integrity checks; AstroM3Edge remains a future integration.

17.6 Phase 5: Dynamic degradation

Implement sustained-violation detection, hysteresis, safe switching, memory overlap handling, epoch validation, and observation expiry. Acceptance: no model oscillation under transient load, no mixing of stale results across epochs, and no unbounded native work after a timeout. Opportunistic promotion during a critical observing transaction remains excluded.

A Technology Stack and Source Authorities

Area	Authority
Daemon	Rust/C++ bridge and `achernar-daemon/src`; `Cargo.toml` and lockfile determine actual dependencies.
Client/desktop	`achernar-frontend` and `achernar-desktop`; package manifests and runtime transport code define current behavior.
Backend	`backend`; installed applications and URL configuration define live services, not legacy architecture inventories.
ML	`ml/README.md`, contracts, datasets, training, exporters, evaluation and artifact metadata.
Schemas	`achernar-daemon/migrations`; architecture examples do not replace migrations.
Runtime ADR	`docs/adr/compute-aware-model-runtime.md`. Detailed implementation decision contract for this revision.
Document guide	`docs/ARCHITECTURE_DOCUMENTS.md`. Source, generation, review, and revision policy.

B Deployment Decisions and Trade-offs

Decision	Rationale and cost
Finite independent variants first	Straightforward artifact identity, calibration, debugging, rollback, and quality comparison; extra training cost.
Measured candidate qualification	Captures provider, precision, batch, and device effects; requires benchmark tooling and hardware access.
Classical baseline	Maintains offline operation without trained artifacts; sets a measurable quality baseline.
Session-pinned initial selection	Simplifies reproducibility and focus-curve consistency; does not automatically exploit transient spare compute.
Separate control algorithms	Avoids maintaining unnecessary GPU tiers for small trees, filters, bandits, and optimizers.

The broad v4 comparison claiming traditional tools lack offline operation is not retained: this specification does not establish such a market comparison. Product claims should be supported by a separate current comparative study.

C Future Research

Elastic/slimmable networks, hardware-aware architecture search, self-supervised astronomy pretraining, uncertainty ensembles, temporal COC, and richer local embeddings remain research directions. Investigate them only when simpler variants expose a measurable limitation. Reinforcement learning remains in simulation until replay, interlocks, and hardware evidence justify more.

D Hardware Validation Matrix

Platform	Candidate path	Required qualification
Raspberry Pi-class Linux	CPU, possibly INT8	aarch64 build, operator support, actual model quality/latency, RAM and thermal headroom.
Original Jetson Nano	Compatible aarch64/NVIDIA stack	Device-specific runtime, power mode, memory, precision/operator support; no automatic Large tier.
Jetson Orin family	Separately qualified NVIDIA path	Independent device and software fingerprint; no inherited Nano result.
Linux NVIDIA laptop/workstation	CUDA and optionally TensorRT	Driver/provider package, actual placement, transfer cost, stacking/display co-load.
Windows laptop	CPU initially; DirectML candidate	Build, supported operators, parity, device memory, concurrent display workload.
Apple Silicon	CPU initially; CoreML candidate	Backend/operator mapping, unified-memory accounting, sustained and concurrent load.
Intel mini PC	CPU; OpenVINO candidate	Exact CPU/accelerator path, threading, parity and measured resource envelope.

This is a qualification plan, not a list of tested releases or model-size guarantees. Published generic benchmarks cannot supply Achernar-specific latency, memory, accuracy, or safe operating thresholds.

E Traceability to v4

The original PDF and its extracted text are retained unchanged by this revision. V5 preserves the numbered top-level subject areas while consolidating old code/SQL sketches into requirements and source references. Exact historical listings remain in v4; do not treat them as current executable examples.

v4 area	v6 disposition (v5 baseline retained)
1–3	Retains purpose, layers, target users, equipment; marks capabilities as implemented or planned rather than claiming universal support.
4.3, 6.7	Replaces one compact local model with family/artifact/candidate selection, orthogonal states, qualification, cache and fallback requirements.
5, 7.4–7.5	Points to current migrations/router; distinguishes legacy proposed routes from current routes and proposed runtime introspection.
6.2, 7.6.1	Reconciles FQE with per-star 64-pixel batching, log-HFR pixels, no position leakage, calibrated pooling, and proposed scalable variants.
7.6.2–3	Preserves FOP features with executable contract; removes unjustified mandatory distillation; updates FDF to the current three-state filter.
7.6.4–10	Retains COC/TED/BGE and other families; distinguishes neural placeholders, fixed CPU algorithms, embedding compatibility, and manifest v2.
7.7–7.11	Preserves grouped data, replay, shadow and bounded autonomy; adds deployed parity, task-specific metrics, slice gates, and runtime qualification.
8	Preserves shared observations, filter handshake, and all FW/M models; corrects physical units, feature-dimension assumptions, CP-SAT sketch, and pointing geometry cautions.
9–10	Retains data and metric topics; adds label provenance, coverage, independent quality evidence, and complete pipeline measurements.
11–12	Replaces uniform INT8/sub-50 ms local framing with qualified providers, per-subsystem budgets, and local placement independent of capacity.
13–14	Retains network/client/cloud responsibilities; avoids claiming unimplemented transport or latency guarantees.
15–16	Replaces “make all edge models tiny” with resource-qualified variants and evidence-gated phases instead of fixed-week promises.
A–D	Retains stack, decisions, research and hardware topics; removes unsupported market comparisons and turns hardware claims into validation plans.

F Implementation and Documentation Map

Target	Future responsibility or documentation change
`models/manifest.rs`	Typed v2 bundle, contracts, hashes and lifecycle.
`models/model_registry.rs`	Complete artifact inventory and legacy adapter; filter eligibility before selection.
`models/execution_provider.rs`	Compiled-provider discovery, placement, session probe, options, and correctness.
`models/compute_profile.rs`	Advisory inventory and current resource budget.
`models/benchmark.rs`	Measurement, fingerprints, cache, and invalidation.
`models/selector.rs`, `runtime.rs`	Qualification policy, shared reservations, family handles, health and baseline.
`models/onnx_runner.rs`	Implement or replace unfinished generic runner; avoid keeping a second unused abstraction.
`vision/fqe_runner.rs`, `preprocess.rs`	Versioned family adapter; preserve batching, calibration, aggregation and physical units.
`vision/vision_loop.rs`, `state.rs`	Bounded workers, handles, frame/epoch validity and classical authority.
`config.rs`, `api/ai.rs`	Future policy and introspection; versioned config migration and admin controls.
`ml/export`, `ml/eval`	Manifest, golden vectors, precision/provider parity, quality and promotion evidence.
README and ML README	Current/experimental/planned status and artifact lifecycle.
Build and testing guides	Provider release matrix, representative benchmarking, parity, cache, failure and resource tests.
Streaming architecture	Current routes/fields, shared observations, future queue and selection provenance.
Dated audit addendum	Current gaps and accepted changes without rewriting earlier audit findings.

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